So MPC physical plant Linearized: $g = 9.8 \, m/s^2$

$$x_{i+1} = A_i x_i + B_i u_i + G_i \vec{g}$$

where $\vec{x_i} = \begin{bmatrix} R_i \\ P_i \\ \omega_i \\ v_i \end{bmatrix}$. $\vec{u_i} = \begin{bmatrix} \vec{f_1} \\ \vec{f_2} \\ \vec{f_3} \\ \vec{f_4} \end{bmatrix}$, $\vec{g} = \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix}_{4} = -g[1\ 1\ 1\ 1]$.

$$\vec{A_i} = \begin{bmatrix} 1 & 0 & R_z(\psi)_i^T \Delta t_i & 0 \\ 0 & 1 & 0 & \Delta t_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$\vec{B_i} = \begin{bmatrix} \frac{1}{2}(\Delta t_i)^2 \cdot \vec{B}_{w_i} \\ \frac{1}{2}(\Delta t_i)^2 \cdot \vec{B}_{v_i} \\ \Delta t_i \, \vec{B}_{w_i} \\ \Delta t_i \, \vec{B}_{v_i} \end{bmatrix}.$$

where $\begin{cases} \vec{B}_{w_i} = I^{-1} \cdot \begin{bmatrix} 0 & -r_{z_1} & r_{y_1} & | & 0 & \cdots & r_{y_4} \\ r_{z_1} & 0 & -r_{x_1} & | & r_{z_2} & \cdots & -r_{x_4} \\ -r_{y_1} & r_{x_1} & 0 & | & -r_{y_2} & \cdots & 0 \end{bmatrix} \\ \vec{B}_v = \frac{1}{m}[1\ 1\ 1\ 1]. \end{cases}$

$$G_i = \begin{bmatrix} 0 \\ \frac{1}{2}\Delta t_i^2 \\ 0 \\ \frac{1}{2}\Delta t_i^2 \end{bmatrix}_{4\times 1} [-g \ -g \ -g \ -g]_{1\times 4}.$$

Loss function: $J = \sum_{i=0}^{N} \left[(\vec{x_i} - \vec{x}_{i_{ref}})^T \vec{Q} \cdot (\vec{x_i} - \vec{x}_{i_{ref}}) + (\vec{u_i} - \vec{u}_{i_{ref}})^T R (\vec{u_i} - \vec{u}_{i_{ref}}) \right]$

constraints: $\begin{cases} ① |f_{i_3}| \leq F_{max.} \approx \mu F_N \approx \frac{\mu}{2} mg. \quad \text{(stance)}. \\ ② f_{swing} = 0. \quad ; \quad ③ \left| J_{i,B}(\vec{q}_i)^T \cdot R_{WB}^T \cdot f_{iw} \right| \leq (\tau_{max}) \end{cases}$

MPC:

States: $X = \begin{bmatrix} R \\ P \\ w \\ v \end{bmatrix} \in R^{12}$.

$R = [\phi, \theta, \psi]^T$, roll, pitch, yaw.

$P =$ COM position.

$w =$ angular vel

$V =$ COM vel (translational).

controller input: $u = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix} \in R^{12}$.

Physic Plant:

$$\begin{cases} m\vec{a} = \sum_i f_i + f_g \\ I\vec{\alpha} + \vec{w} \times I\vec{w} = \sum_i r_i \times f_i + \tau_g \end{cases}$$

$$\begin{cases} \vec{a} = \frac{1}{m}\sum_i f_i + F_g \\ \vec{\alpha} \approx I^{-1}\left(\sum_i \vec{r_i} \times \vec{f_i} + \vec{\tau_g}\right) \end{cases}$$

if nearly flat, $\vec{w}$ small.
neglect $\vec{w} \times I\vec{w}$.

$\dot{R} = R_z(\psi)^T \cdot \vec{w}$ if ϕ, θ small.

$X_{i+1} = A_i X_i + B_i u_i + G_i$ $\quad G_i = \begin{bmatrix} f_g \\ \tau_g \end{bmatrix}$

$$X_i = \begin{bmatrix} \phi & P_x & w_x & v_x \\ \theta & P_y & w_y & v_y \\ \psi & P_z & w_z & v_z \end{bmatrix}_i \quad 12\times 1 \text{ column matrix.}$$

$$\begin{cases} w_{i+1} = w_i + \alpha_i \Delta t. \\ V_{i+1} = V_i + a_i \Delta t \\ P_{i+1} = P_i + V_i \Delta t + \frac{1}{2} a_i (\Delta t)^2 \\ R_{i+1} = R_z(\psi)_i^T \cdot \vec{w}_i \Delta t. \end{cases}$$

$$X_{i+1} = \begin{bmatrix} \phi \\ \theta \\ \psi \\ P_x \\ P_y \\ P_z \\ w_x \\ w_y \\ w_z \\ v_x \\ v_y \\ v_z \end{bmatrix}_{i+1} = \begin{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} R_{i+1} \\ P_{i+1} \\ w_{i+1} \\ V_{i+1} \end{bmatrix}_{4\times 1} = \begin{bmatrix} R_i + R_z(\psi)_i^T \cdot w_i \Delta t_i \\ P_i + V_i \Delta t_i + \frac{1}{2} a_i (\Delta t_i)^2 \\ w_i + \alpha_i \Delta t_i \\ V_i + \alpha_i \Delta t_i \end{bmatrix}$$

$$= \begin{bmatrix} R_i & P_i & w_i & V_i \\ I & 0 & R_z(\psi)_i^T \Delta t_i & 0 \\ 0 & I & & \\ & & I & \\ & & & I \end{bmatrix}_{4\times 12} \begin{bmatrix} R_i \\ P_i \\ w_i \\ V_i \end{bmatrix}_{12\times 1}$$

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \end{bmatrix}_{2\times3} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}_{3\times1} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}_{2\times1}$$

1+0+3 = 0

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}_{1\times4} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}_{4\times1}$$

A:

$$\begin{bmatrix} 1 & 0 & R_2(\psi)_i^T \Delta t_i & 0 \\ 0 & 1 & 0 & \Delta t_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_i \\ P_i \\ \omega_i \\ v_i \end{bmatrix}$$

$+\frac{1}{2}\alpha_i^2(\Delta t_i)^2$

$+\omega_i \Rightarrow \omega_i \times + \alpha$

$$= \begin{bmatrix} R_i + R_2(\psi)_i^T \Delta t_i \omega_i \\ P_i + v_i \Delta t_i \\ \omega_i \\ v_i \end{bmatrix}$$

$$= \frac{1}{m} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}$$

$$a_i = \frac{1}{m}(f_1+f_2+f_3+f_4) \pm \frac{1}{m} f_1 \pm \frac{1}{m} f_2$$

$$a_i = \frac{1}{m}[1\ 1\ 1\ 1] \cdot \vec{u}$$

$$\vec{u} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}$$

$$\begin{cases} \vec{a}_i = \frac{1}{m} \sum_i^4 f_i \pm \vec{g} \\ \vec{\alpha}_i = I^{-1} \sum_i^4 \vec{r}_i \times \vec{f}_i \pm \vec{\alpha} \end{cases}$$

$B\vec{u}$:

$$X_{i+1} = \begin{bmatrix} R_i + R_2(\psi)_i^T \omega_i \Delta t_i \\ P_i + v_i \Delta t_i \\ \omega_i \\ V_i \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2} a_i (\Delta t_i)^2 \\ \alpha \cdot d_i (\Delta t_i) \\ a_i \Delta t_i \end{bmatrix}$$

$$B\vec{u} = \begin{bmatrix} 0 \\ \frac{1}{2}(\Delta t_i)^2 a_i \\ \alpha_i \Delta t_i \alpha_i \\ a_i \Delta t_i \end{bmatrix} =$$

$$\vec{r_i} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix}_{12\times 1} \qquad \vec{u} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}_{12\times 1}$$

$r_i \times f_i$ ⇒ $r_i \times (\vec{u})$

$$\vec{r_i} = \begin{bmatrix} r_{xi} \\ r_{yi} \\ r_{zi} \end{bmatrix} \qquad \vec{f_i} = \begin{bmatrix} f_{xi} \\ f_{yi} \\ f_{zi} \end{bmatrix}$$

$$r_i \times f_i = \begin{vmatrix} i & j & k \\ r_{xi} & r_{yi} & r_{zi} \\ f_{xi} & f_{yi} & f_{zi} \end{vmatrix} = i\begin{vmatrix} r_{yi} & r_{zi} \\ f_{yi} & f_{zi} \end{vmatrix} - j\begin{vmatrix} r_{xi} & r_{zi} \\ f_{xi} & f_{zi} \end{vmatrix} + k\begin{vmatrix} r_{xi} & r_{yi} \\ f_{xi} & f_{yi} \end{vmatrix}$$

$$= \begin{vmatrix} r_{yi} f_{zi} - r_{zi} f_{yi} \\ r_{zi} f_{xi} - r_{xi} f_{zi} \\ r_{xi} f_{yi} - r_{yi} f_{xi} \end{vmatrix}$$

$$= \begin{bmatrix} 0 & -r_{zi} & r_{yi} \\ r_{zi} & 0 & -r_{xi} \\ -r_{yi} & r_{xi} & 0 \end{bmatrix} \begin{bmatrix} f_{xi} \\ f_{yi} \\ f_{zi} \end{bmatrix}$$

$$\vec{G} = \begin{bmatrix} F_g \\ \tau_g \end{bmatrix}$$

So $\tau_i = \begin{bmatrix} 0 & -r_{zi} & r_{yi} \\ r_{zi} & 0 & -r_{xi} \\ -r_{yi} & r_{xi} & 0 \end{bmatrix} \cdot \begin{bmatrix} f_{xi} \\ f_{yi} \\ f_{zi} \end{bmatrix}$.

So $\tau(\text{total}) = \sum_{i=1}^{4} \tau_i = I^T \begin{bmatrix} 0 & -r_{z1} & r_{y1} & 0 & \cdots \\ r_{z1} & 0 & -r_{x1} & r_{z2} & \cdots \\ -r_{y1} & r_{x1} & 0 & -r_{y2} & \cdots \end{bmatrix}_{3\times 12} \cdot \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}_{12\times 1}$.

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}a\\ b\end{bmatrix}=\begin{bmatrix}a\\ b\end{bmatrix}.$$

$$\begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix}_{1\times 4}\cdot\begin{bmatrix}f_1\\ f_2\\ f_3\\ f_4\end{bmatrix}_{4\times 1}$$

$$\begin{bmatrix}1 & 0 & 3\\ 2 & 1 & 2\end{bmatrix}_{2\times 3}\cdot\begin{bmatrix}1\\ 1\\ 1\end{bmatrix}_{3\times 1}=\begin{bmatrix}4\\ 5\end{bmatrix}_{2\times 1}.$$

$1+0+3=0$

A:

$$\begin{bmatrix}1 & 0 & R_2(\psi)_i^T\Delta t_i & 0\\ 0 & 1 & 0 & \Delta t_i\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}R_i\\ P_i\\ \omega_i\\ v_i\end{bmatrix}$$

$+\frac{1}{2}\alpha_i(\Delta t_i)^2$

$+\omega_i=\omega_i+\alpha$

$$=\begin{bmatrix}R_i+R_2(\psi)_i^T\Delta t_i\,\omega_i\\ P_i+v_i\Delta t_i\\ \omega_i\\ v_i\end{bmatrix}$$

$$=\frac{1}{m}\cdot\begin{bmatrix}1 & 1 & 1 & 1\\ 0 & 0 & 1 & 1\\ 0 & 0 & 1 & 1\end{bmatrix}\cdot\begin{bmatrix}f_1\\ f_2\\ f_3\\ f_4\end{bmatrix}$$

$$a_i=\frac{1}{m}(f_1+f_2+f_3+f_4)\quad \cancel{\frac{1}{m}f_1+\frac{1}{m}f_2}$$

$$a_i=\frac{1}{m}[1\ 1\ 1\ 1]\cdot\vec{u}$$

BU:

$$\vec{u}=\begin{bmatrix}f_1\\ f_2\\ f_3\\ f_4\end{bmatrix}$$

$$\begin{cases}\vec{a}_i=\frac{1}{m}\sum_i^4 f_i+\cancel{\vec{g}}\\ \vec{\alpha}_i=I^{-1}\sum_i^4\vec{r}_i\times\vec{f}_i+\cancel{\vec{g}}\end{cases}$$

$$X_{i+1}=\begin{bmatrix}R_i+R_2(\psi)_i^T\omega_i\Delta t_i\\ P_i+v_i\Delta t_i\\ \omega_i\\ v_i\end{bmatrix}+\begin{bmatrix}0\\ \frac{1}{2}a_i(\Delta t_i)^2\\ \alpha\cdot d_t(\Delta t_i)\\ d_t a_i\Delta t_i\end{bmatrix}$$

$$BU=\begin{bmatrix}0\\ \frac{1}{2}(\Delta t_i)^2 a_i\\ d_t\Delta t_i\,\alpha_i\\ a_i\Delta t_i\end{bmatrix}=$$

So $\vec{X}_{i+1} = \vec{A} \cdot \vec{X}_i + \vec{B} \cdot u_i +$

$\vec{G} = \begin{bmatrix} F_g \\ \tau_g \end{bmatrix} = \begin{bmatrix} F_{gx} \\ F_{gy} \\ F_{gz} \\ \tau_{gx} \\ \tau_{gy} \\ \tau_{gz} \end{bmatrix}$.

$\vec{B}\vec{u} = \begin{bmatrix} 0 \\ \frac{1}{2} a_i (\Delta t_i)^2 \\ a_i \Delta t_i \\ a_i \Delta t_i \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

$\vec{g} = \begin{bmatrix} 0 \\ 0 \\ -9.8 \end{bmatrix}$

$m = \text{const}$

$m\vec{g} \rightarrow \vec{a}_g = \cdot m\vec{g}$

$\vec{a}\vec{d}_g = \vec{P}_i \times \vec{g}$

$$\begin{cases} a_i = \frac{1}{m} [1\ 1\ 1\ 1]\ \vec{u} + \\ \vec{a}\vec{d}_i = \vec{T} \cdot \vec{u} + \vec{P}_i \times \vec{g} \end{cases}$$

$\vec{B} = \begin{bmatrix} 0 \\ \frac{1}{2}(\Delta t_i)^2 \cdot \frac{1}{m} [1\ 1\ 1\ 1]\ \vec{u} \\ (\Delta t_i)\ \vec{T}\vec{u} \\ (\Delta t_i)\ \frac{1}{m} [1\ 1\ 1\ 1]\ \vec{u} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(\Delta t_i)^2 \cdot \vec{T}a \\ \frac{1}{2}(\Delta t_i)^2 \cdot \frac{1}{m} [1\ 1\ 1\ 1] \\ \Delta t_i\ \vec{T} \\ \Delta t_i\ \frac{1}{m} [1\ 1\ 1\ 1] \end{bmatrix} \cdot \vec{u}$

$\vec{G} = \begin{bmatrix} I^{-1} \tau_g \\ -mg \end{bmatrix}$

$R_{i+1} = R_i + R_z(\alpha)_i^T w_i \Delta t_i + \frac{1}{2} \alpha_i (\Delta t_i)^2 + \frac{1}{2} \alpha_{gi} (\Delta t_i)^2$

$\vec{C} \cdot \vec{G}$

$G = \vec{g}$

$\vec{G} = \begin{bmatrix} F_g \\ \tau_g \end{bmatrix}$

q_{k+1}

$\text{Pos}_{i+1} = \cdots\cdots\cdots + \begin{bmatrix} + \frac{1}{2} a_{gi} (\Delta t_i)^2 \\ + a_{gi} \Delta t_i \\ + a_{gi} \Delta t_i \end{bmatrix}$

$w_{i+1} = \cdots\cdots\cdots$

$v_{i+1} = \cdots\cdots\cdots$

$= \begin{bmatrix} \frac{1}{2} (\Delta t_i)^2 \cdot$

$$\vec{\tau}_{g_i} = \vec{P_i} \times \vec{g} = \begin{vmatrix} i & j & k \\ P_{xi} & P_{yi} & P_{zi} \\ 0 & 0 & -g \end{vmatrix} \qquad g = 9.8 \text{ m/s}^2$$

$$= i\begin{vmatrix} P_{yi} & P_{zi} \\ 0 & -g \end{vmatrix} - j\begin{vmatrix} P_{xi} & P_{zi} \\ 0 & -g \end{vmatrix} + k\begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

$$= \begin{bmatrix} -P_{yi}\, g \\ P_{xi}\, g \\ 0 \end{bmatrix} = \begin{bmatrix} -P_{yi} \\ P_{xi} \\ 0 \end{bmatrix} \cdot \vec{g} \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} =$$

So $\alpha_{g_i} = I^{-1} \cdot \begin{bmatrix} -P_{yi} \\ P_{xi} \\ 0 \end{bmatrix} \cdot \vec{g}$

$$a_{g_i} = \vec{g} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \vec{g}$$

$$\begin{bmatrix} \frac{1}{2}\alpha_{g_i}(\Delta t_i)^2 \\ \frac{1}{2} a_{g_i}(\Delta t_i)^2 \\ \alpha_{g_i}(\Delta t_i) \\ a_{g_i}(\Delta t_i) \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}\Delta t^2 I^{-1} \begin{bmatrix} -P_{yi} \\ P_{xi} \\ 0 \end{bmatrix} \\ \frac{1}{2} a_{g_i} \Delta t_i \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\ a\, \Delta t_i\, I^{-1} \begin{pmatrix} P_{yi} \\ P_{xi} \\ 0 \end{pmatrix} \\ a\, \Delta t_i \cdot I_3 \end{bmatrix} \vec{g}$$

if origin is at CoM.

then $\vec{G} = \begin{bmatrix} 0 \\ \frac{1}{2} a \Delta t_i^2 \\ 0 \\ \frac{1}{2} \Delta t_i^2 \end{bmatrix} \cdot \vec{g}$.